

llmXive Automated Review of The Topological Trouble With Transformers

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer’s exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer’s section also names the underlying model that produced it.

This paper’s panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The manuscript makes several strong theoretical claims regarding the expressivity and state-

tracking capabilities of various architectures, specifically contrasting feedforward transformers with recurrent and state-space models. While the overall argument is coherent, specific citations require verification to ensure they support the precise claims made.

In Section 2, the paper cites [baldelli2026](#) to formally address the failure of models to maintain consistent hidden states. Given that this is a 2026 preprint, it is critical to verify that the paper explicitly contains the formal proof or analysis attributed to it, rather than just anecdotal evidence. Similarly, the claim that “SSMs with linear updates are no more expressive than an ordinary transformer” (Section 5) relies heavily on [merrill2025illusion](#). The review must confirm that this source indeed establishes a strict equivalence in expressivity that precludes linear SSMs from solving state-tracking problems that transformers cannot, or if the distinction is more nuanced (e.g., regarding efficiency or specific inductive biases).

Furthermore, the detailed mechanistic explanation of the “bank” ambiguity example (Section 2), specifically the claim that disambiguation occurs at the sixth block and is inaccessible to shallower layers, is attributed to [lepori2025](#). The authors must ensure that [lepori2025](#) explicitly reports this specific layer-depth finding for this specific example, as opposed to a general observation of racing thoughts. If the cited paper only demonstrates the error without the specific layer-wise analysis, the claim is overstated.

Finally, the assertion in Section 3 that “depth recurrence... does not enable indefinite state tracking” is a central theoretical pillar. The citation [merrill2025](#) (or [merrill2025illusion](#)) must be checked to ensure it specifically addresses the limitations of *depth* recurrence (looped transformers) versus *step* recurrence, and that the proof holds for the specific definition of “indefinite tracking” used in the text. If the cited work only bounds the depth required for specific languages (like regular languages) without addressing the general state-tracking capacity of looped architectures, the claim may be too broad.

Action items.

- **[science]** Verify the claim that ‘SSMs with linear updates are no more expressive than an ordinary transformer’ (Section 5) against the cited source [Merrill2025illusion](#). Ensure the citation does not overstate the equivalence, as some linear SSMs may have distinct inductive biases or efficiency profiles not captured by pure expressivity bounds.
- **[science]** Confirm that the citation ‘[baldelli2026](#)’ (Section 2, footnote) actually contains the formal proof regarding ‘failure of standard models to reliably maintain a consistent hidden state’ as attributed. The bibliography lists this as a 2026 preprint; verify the specific claim exists in the referenced text.
- **[science]** Check the attribution of the ‘bank’ ambiguity example to ‘[lepori2025](#)’ (Section 2). Ensure the paper explicitly demonstrates the ‘racing thoughts’ mechanism (disambiguation at layer 6 vs. shallow layers) for this specific example, rather than just reporting the error generally.
- **[science]** Validate the claim that ‘depth recurrence... does not enable indefinite state tracking’ (Section 3) against the cited ‘[merrill2025](#)’ or ‘[merrill2025illusion](#)’. Ensure the distinction between ‘depth recurrence’ (looped) and ‘step recurrence’ is clearly supported by the cited theoretical bounds.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 provides a clear schematic of the transformer architecture, but the caption fails to define the specific symbols used in the diagrams, such as the purple and green arrows in panel (b) and the specific connectivity lines in panel (a).

Figure 2

Figure 2 illustrates the upward flow of information in two external models as described in the caption, but the x-axis text labels are illegible and the y-axis lacks specific layer indices to quantify the depth.

Figure 3

The figure effectively visualizes the concept of depth-limited inference with clear icons and flow arrows, but the caption contains a critical truncation error at the end and slightly confusing phrasing regarding the spatial layout of inputs versus layers.

Figure 4

The figure conceptually illustrates the unrolling of a recurrent network as described in the caption, but the image quality is poor. Specifically, the weight labels are illegible and the descriptive text labels are partially cropped.

Figure 5

The figure is visually clear but the caption is truncated, failing to define panel (d) and cutting off the description of panel (c), which renders the figure's full content unintelligible.

Figure 6

The figure is a blank grid that fails to visualize the described latent-thought model; it lacks the blue input token marker mentioned in the caption and contains no labels, arrows, or legends to explain the architecture.

Figure 7

The rendered image for Figure 7 appears to be a duplicate of Figure 1(b) and does not depict the SSM unrolling described in the caption, rendering the figure scientifically invalid for its stated purpose.

Action items.

- **[writing]** Figure 1: The caption for panel (b) mentions ‘green rectangles’ but does not explain the meaning of the purple arrows or the green curved arrows connecting the blocks.
- **[writing]** Figure 1: The caption for panel (a) describes ‘colored lines’ but does not define the specific meaning of the red, blue, and yellow connections shown.
- **[writing]** Figure 2: The x-axis labels (e.g., ‘Feed’, ‘the’, ‘bank’) are illegible due to low resolution and rotation; they should be enlarged or reoriented for readability.
- **[writing]** Figure 2: The y-axis label ‘blocks’ is present, but the specific block numbers or layer indices corresponding to the grid lines are missing, making it impossible to quantify the ‘upward flow’ mentioned in the caption.
- **[fatal]** Figure 3: The caption ends abruptly with ‘resulting in an incorrect prediction at the ‘ and is missing the final word (likely ‘ATM’) and the closing bracket for the citation.
- **[science]** Figure 3: The caption states ‘input tokens are presented at the bottom’, but the image displays the input sequence (day off work, fishing pole, etc.) along the bottom axis with layers stacked vertically above; the caption should clarify that layers are processed upwards from the bottom.
- **[writing]** Figure 4: The labels ‘w1’, ‘w2’, ‘w3’, and ‘w4’ are rendered with extremely low resolution and are illegible, making it impossible to verify the specific weight connections described in the caption.
- **[writing]** Figure 4: The text labels ‘recurrent network’ and ‘unrolled (feedforward) network’ are cut off on the left and right edges respectively, reducing clarity.
- **[fatal]** Figure 5: The caption ends abruptly with ‘(c) An unrolled bloc’, indicating missing text that likely defines the content of panel (c) and potentially panel (d), which is visible in the image but undefined.
- **[science]** Figure 5: The image displays four panels labeled (a) through (d), but the caption only describes (a), (b), and a truncated (c), leaving panel (d) completely undefined.
- **[fatal]** Figure 6: The caption states the next input token is ‘marked with the blue color’, but the rendered image contains no blue elements, making the figure impossible to interpret.
- **[fatal]** Figure 6: The image is completely devoid of text labels, axis titles, or a legend, rendering the schematic meaningless without external context.
- **[fatal]** Figure 6: The caption references ‘multiple auto-regressive steps’ and ‘latent thoughts’, but the grid contains no arrows or indicators to show the direction of flow or feedback loops.

- **[fatal]** Figure 7: The rendered image displays a schematic of a transformer decoder (labeled ‘a’ and ‘b’) with colored input tokens and state propagation arrows, which matches the description of Figure 1(b) rather than the caption’s description of an SSM unrolling with horizontal flow.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript exhibits a high density of specialized terminology that, while standard within the specific niche of recurrent transformer research, risks alienating a broader audience of cognitive scientists and general machine learning practitioners. The primary issue is the introduction of terms like “autoregressive unrolling” (Section 3) and “canon layers” (Section 5) without immediate, explicit definitions or sufficient context for a non-specialist reader. While “attractor dynamics” and “arithmetic intensity” are defined in bolded blocks, these definitions are somewhat isolated from the narrative flow, requiring the reader to pause and cross-reference.

Specifically, the term “autoregressive unrolling” is central to the paper’s argument about the necessity of sequential dependency, yet the distinction made between this and standard token generation is subtle and relies on the reader already understanding the nuances of training vs. inference modes in recurrent architectures. A clearer, more explicit definition at the point of first use would significantly improve accessibility. Similarly, “canon layers” is used as a key example of leveraging representational alignment but is not defined, forcing the reader to guess its meaning or search external literature.

Additionally, the use of colloquialisms like “flip-flop” (Section 2) to describe state tracking failures, while vivid, detracts from the formal tone expected in a theoretical analysis. Replacing this with a more precise technical description would align the prose better with the paper’s rigorous arguments. The definitions provided for “teacher forcing” and “attractor dynamics” are helpful but should be integrated more smoothly into the text rather than presented as isolated glossary entries. Overall, the paper would benefit from a “jargon audit” to ensure every technical term is either defined upon first use or replaced with a more accessible equivalent.

Action items.

- **[writing]** The manuscript exhibits a high density of specialized terminology that, while standard within the specific niche of recurrent transformer research, risks alienating a broader audience of cognitive scientists and general machine learning practitioners. The primary issue is the introduction of terms like “autoregressive unrolling” (Section 3) and “canon layers” (Section 5) without immediate, explicit definitions or sufficient context for a non-specialist reader. While “attractor dynamics” and “ari

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The manuscript presents a coherent high-level argument that the feedforward topology of transformers creates a “depth bottleneck” for state tracking, necessitating recurrent mechanisms. However, several internal logical gaps weaken the causal claims regarding specific architectural classes.

First, in Section 3, the distinction between “depth recurrence” (looped transformers) and “feedforward depth” is muddled. The paper argues that depth recurrence fails to enable indefinite tracking because the state representation “shifts upward” (Section 3, paragraph 4). This contradicts the definition of recurrence provided earlier, where a layer is re-used. If a layer L updates its own state $s_t = f(s_{t-1}, x_t)$, the state does not necessarily shift to a deeper layer $L + 1$ in the unrolled graph; it remains in L . The claim that the state *must* shift upward appears to conflate depth recurrence with the standard feedforward propagation of a single pass. The argument requires a precise mechanistic explanation of why the state in a looped transformer cannot be maintained in a fixed layer without “running out of depth,” or the claim must be qualified to exclude specific looped architectures that successfully maintain state.

Second, there is a logical tension in Section 4 regarding State-Space Models (SSMs). The text states that “SSMs with linear updates are no more expressive than an ordinary transformer,” citing Merrill (2025). Immediately after, it claims DeltaNet (which relies on a linear delta rule) achieves “greater expressivity” when eigenvalues are negative. The paper fails to explicitly bridge this gap: does the negative eigenvalue extension technically violate the “linear update” constraint of the cited theorem, or is the initial claim about SSMs an overgeneralization? Without clarifying why the negative eigenvalue case escapes the expressivity bound, the argument that DeltaNet is superior appears to contradict the preceding theoretical claim.

Finally, the “bank” example in Section 2 is used to demonstrate a fundamental failure of feedforward models. However, the text acknowledges that for certain functions, the state updates can be composed into a single-step function g (Section 2, paragraph 5). The paper does not logically explain why the “bank” disambiguation task *cannot* be solved by such a composition, thereby rendering the failure a limitation of the specific model’s training rather than the architecture’s topology. The conclusion that feedforward models “fundamentally limit” state tracking is too strong given the acknowledged existence of compositional shortcuts.

Action items.

- **[science]** Section 3 claims depth recurrence fails because state ‘shifts upward,’ but re-using a layer implies state stays in place. Clarify why looped transformers cannot maintain state without depth exhaustion.
- **[science]** Section 4 states linear SSMs lack expressivity, then claims DeltaNet (linear) gains it with negative eigenvalues. Explicitly explain how negative eigenvalues bypass the cited

theoretical bound.

- **[science]** Section 2 uses the ‘bank’ failure to argue feedforward limits, yet admits compositional shortcuts exist. Explain why this specific task cannot be solved by such shortcuts to support the ‘fundamental limit’ claim.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The manuscript makes several compelling arguments regarding the topological limitations of feedforward transformers for state tracking. However, the review identifies specific instances where the authors extrapolate beyond the strict bounds of their theoretical arguments or the cited literature, particularly regarding the capabilities of State-Space Models (SSMs) and the necessity of recurrence.

First, in Section 3, the authors state: “While depth recurrence can increase the expressivity of a transformer, it does not enable indefinite state tracking.” This is a strong universal claim. While it holds for standard looped transformers with a fixed number of unrollings, the paper’s own taxonomy in Table 1 includes “Deep Equilibrium Models (DEQ)” in the cell for Depth Recurrence with Ratio = 1. DEQs theoretically solve for a fixed point, effectively simulating infinite depth with a finite number of parameters. By categorically dismissing depth recurrence as insufficient for indefinite tracking without explicitly excluding fixed-point methods, the authors overreach their theoretical scope. The text should be refined to specify that *finite-depth* recurrence is limited, whereas fixed-point formulations may bypass this.

Second, there is a tension in the claims regarding SSMs. In Section 4, the authors assert: “State-space models (SSMs) are often touted as a means of state propagation, but SSMs with linear updates are no more expressive than an ordinary transformer.” This is a precise theoretical claim supported by Merrill (2025). However, in Section 5.1 (“Enhanced State-Space Models”), the text claims that DeltaNet and RWKV-7 “achieve state tracking beyond the capability of ordinary transformers.” While these models are indeed more expressive, the transition from the general dismissal of SSMs to the specific praise of gated/non-linear variants is abrupt. The manuscript risks overclaiming the limitations of the *entire* SSM class by initially grouping them with linear models, then later highlighting non-linear variants as exceptions without a clear demarcation in the initial critique. The distinction between linear and non-linear/gated SSMs must be sharper to avoid misleading the reader about the expressivity of the broader class.

Finally, the conclusion (Section 6) makes a prescriptive claim: “The next generation of foundation models must... maintain a fluid, evolving representation of reality.” This implies that recurrence is the *only* viable path forward. While the paper argues strongly for recurrence, it does not rule out other potential architectural evolutions (e.g., hybrid retrieval mechanisms, dynamic context windows, or novel attention mechanisms) that might solve state tracking

without explicit recurrence. The language should be adjusted to reflect that recurrence is a *promising* and *theoretically sound* direction, rather than a mandatory requirement for the future of foundation models.

These issues represent overreach in the scope of the claims rather than fundamental flaws in the core argument. Addressing them will ensure the paper’s conclusions are strictly supported by the presented evidence and citations.

Action items.

- **[science]** The claim that ‘depth recurrence... does not enable indefinite tracking’ (Section 3) overgeneralizes. While true for standard looped transformers, the paper cites DEQ in Table 1 as a potential candidate for the ‘Depth’ axis, which theoretically allows fixed-point iteration to simulate infinite depth. Clarify that the limitation applies to finite-depth recurrence, not all depth-recurrent formulations.
- **[science]** The assertion that ‘SSMs with linear updates are no more expressive than an ordinary transformer’ (Section 4) is followed by claims that DeltaNet and RWKV-7 achieve ‘state tracking beyond’ transformers (Section 5.1). Reconcile this by explicitly distinguishing between linear SSMs and non-linear/gated variants to avoid overclaiming the limitations of the entire SSM class.
- **[writing]** The conclusion states that ‘the next generation of foundation models must... maintain a fluid, evolving representation’ (Section 6). This prescriptive claim exceeds the paper’s scope, which demonstrates limitations but does not prove recurrence is the only path. Soften the language to ‘a promising direction’ rather than a mandatory requirement.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript addresses critical architectural limitations of transformers regarding state tracking, which has direct implications for AI safety and reliability. However, the presentation of empirical evidence requires greater transparency to ensure the safety analysis is robust and reproducible.

In Section 2, the authors present failure traces from “Gemini 3 (Fast)” and “Gemini 2.5 Flash” to illustrate state tracking deficiencies. While these examples are compelling, the manuscript lacks the necessary metadata to verify these claims. Specifically, the exact model versions, generation parameters (temperature, top_p), and precise prompt sequences are not provided. Without this information, it is difficult to assess whether these failures are inherent to the architecture or artifacts of specific prompting strategies. This lack of reproducibility hinders the community’s ability to rigorously evaluate the safety risks associated with these limitations.

Furthermore, the discussion of “breakdowns in communication and cooperation in multi-agent

settings” (Section 2) touches upon critical safety concerns. The paper should explicitly elaborate on the potential real-world consequences of these architectural failures. For instance, how might inconsistent state tracking in an AI agent lead to harmful outcomes in high-stakes domains such as autonomous driving or medical diagnosis? A brief but explicit discussion on the safety implications of these limitations would strengthen the paper’s contribution to the field of AI safety.

Finally, the references to “Gemini 3” and “Gemini 2.5 Flash” appear to be future-dated or potentially hypothetical. If these models exist, the authors must provide verifiable citations. If they are hypothetical examples intended to illustrate a theoretical point, the text must be clearly labeled as such to prevent readers from drawing incorrect conclusions about the current capabilities and safety profiles of existing AI systems. Clarifying the status of these models is essential for maintaining the integrity of the safety analysis.

Action items.

- **[writing]** Section 2 cites specific model failures (Gemini 3, 2.5 Flash) without disclosing exact versions, parameters, or prompts. Authors must provide a supplementary appendix with full reproducibility details (prompts, API settings) to verify these safety-critical failure modes.
- **[writing]** The paper discusses ‘breakdowns in multi-agent cooperation’ (Section 2) but lacks explicit analysis of real-world safety risks. Authors should briefly discuss potential harmful consequences of state-tracking failures in high-stakes domains like healthcare or finance.
- **[writing]** References to ‘Gemini 3’ and ‘Gemini 2.5 Flash’ appear future-dated. Authors must clarify if these are real models with verifiable citations or hypothetical examples. Misleading claims about model capabilities pose significant safety communication risks.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The manuscript presents a compelling theoretical argument regarding the topological limitations of feedforward transformers for state tracking. However, the scientific evidence supporting the central claims relies disproportionately on qualitative examples and theoretical citations rather than rigorous empirical validation.

In Section 2 (“State tracking”), the authors illustrate failure modes using specific traces from “Gemini 3 (Fast)” and “Gemini 3 Thinking.” While these anecdotes are illustrative, they do not constitute robust scientific evidence. The paper asserts that these failures are “fundamental” to the architecture, yet provides no statistical analysis of failure rates, no control groups (e.g., comparing against recurrent baselines on the same tasks), and no sample size justification. To support the claim that these are systemic architectural flaws rather than training artifacts or

specific model idiosyncrasies, the authors should report aggregated performance metrics (e.g., consistency scores, error rates) across a diverse set of prompts and model sizes.

Furthermore, the distinction made in Section 3 between “depth recurrence” and “step recurrence” is critical to the paper’s thesis. The claim that depth recurrence “does not enable indefinite state tracking” is supported primarily by a reference to \citet{merrill2025} and a logical argument about activation flow. While the logic is sound, the paper would benefit from a direct empirical comparison or a more detailed theoretical derivation showing the specific conditions under which depth recurrence fails where step recurrence succeeds. The taxonomy in Table 1 is a useful conceptual tool, but without accompanying empirical data showing performance differences across the proposed categories, it remains a hypothesis rather than a validated framework.

Finally, the discussion of “racing thoughts” and delayed disambiguation (Section 2, Figure 3) cites \citet{lepori2025}. While this is a valid citation, the current manuscript does not independently verify these findings or extend them with new data. To strengthen the evidence base, the authors should either include their own replication of these findings with clear statistical reporting (effect sizes, confidence intervals) or explicitly frame the discussion as a synthesis of existing literature rather than a presentation of new empirical results. The current mix of theoretical claims and anecdotal evidence leaves the central argument vulnerable to alternative explanations, such as insufficient training data or specific prompt engineering issues, rather than inherent topological constraints.

Action items.

- **[science]** The manuscript relies heavily on anecdotal evidence (e.g., specific Gemini 3 traces in Section 2) to support the claim of fundamental architectural failure. To strengthen scientific evidence, provide quantitative metrics (e.g., error rates, consistency scores) aggregated over a statistically significant sample size ($n > 30$) across multiple model variants, rather than isolated failure cases.
- **[science]** The claim that ‘depth recurrence... does not enable indefinite state tracking’ (Section 3) is a strong theoretical assertion. The paper cites \citet{merrill2025} but lacks a direct empirical demonstration or a formal proof sketch within the text showing why specific depth-recurrent architectures fail on tasks requiring unbounded state accumulation compared to step-recurrent ones.
- **[science]** The taxonomy in Table 1 categorizes architectures based on theoretical properties, but the paper does not present empirical benchmarks comparing the state-tracking performance of models in different cells (e.g., Depth vs. Step recurrence). Without comparative data, the classification remains speculative rather than evidence-based.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The manuscript presents a strong theoretical argument regarding the topological limitations of feedforward transformers for state tracking. However, from a statistical analysis perspective, there is a significant disconnect between the claims of empirical validation and the evidence presented in the text.

Specifically, a comment in `topological_trouble.tex` (lines 13-14) asserts: “NOTE: All claims regarding model failures are empirically validated with statistical aggregation across multiple runs (n=10); see Section 4.2 for quantitative analysis (paired t-tests, ANOVA, Cohen’s d effect sizes) and Table 3 for aggregated metrics.” This statement is problematic for two reasons. First, Section 4.2 (Architectural limitations and workarounds) and the rest of the manuscript do not contain a “Section 4.2” dedicated to quantitative analysis, nor is there a “Table 3” presenting aggregated metrics, p-values, or effect sizes. Second, the primary evidence for state-tracking failures in Section 2 consists of qualitative case studies (e.g., the “bank” polysemy example and the “twenty questions” game trace). While these examples are illustrative, they do not constitute the statistical aggregation described in the comment.

The reliance on single-instance anecdotes to support broad claims about “fundamental limitations” and “failure modes” without accompanying statistical rigor (e.g., confidence intervals, error bars, or significance testing across a dataset) weakens the empirical support for the paper’s central thesis. If the authors have indeed conducted the n=10 runs and statistical tests mentioned in the comment, these results must be explicitly reported in the main text with appropriate statistical measures (means, standard deviations, p-values, effect sizes) to validate the claims. If these analyses were not performed, the specific references to t-tests, ANOVA, and sample sizes in the comments must be removed to prevent misleading the reader regarding the nature of the evidence.

Furthermore, the paper cites external works (e.g., `\citet{lepori2025}`) for quantitative findings but does not provide its own statistical synthesis of the proposed taxonomy or the performance of the suggested recurrent architectures. To meet the standards of statistical analysis, the manuscript should either present the missing quantitative data or reframe the arguments to clearly distinguish between theoretical proofs, illustrative qualitative examples, and empirically validated statistical findings.

Action items.

- **[writing]** The manuscript presents a strong theoretical argument regarding the topological limitations of feedforward transformers for state tracking. However, from a statistical analysis perspective, there is a significant disconnect between the claims of empirical validation and the evidence presented in the text. Specifically, a comment in `topological_trouble.tex` (lines 13-14) asserts: “NOTE: All claims regarding model failures are empirically validated with statistical aggregation across multiple runs

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript presents a compelling argument regarding the topological limitations of transformers for state tracking, and the overall narrative is generally clear. However, there are several instances where sentence structure, historical accuracy, and grammatical completeness hinder the reader’s understanding.

First, the Introduction contains a historically confusing statement: “Until the transformer... came along, feedforward nets did not seem like a viable approach to replicating human thought and reasoning.” This phrasing is misleading because standard feedforward networks (MLPs) existed long before transformers and were used for various tasks. The sentence likely intends to say that *purely feedforward sequence models* (lacking recurrence or attention) were insufficient for long-range dependencies, but the current wording suggests a broader historical inaccuracy that distracts from the main argument.

Second, the flow of the text is occasionally interrupted by abrupt definitions. In Section 3, the definitions of “Teacher forcing” and “Attractor dynamics” appear as a standalone paragraph immediately following the section header. While these terms are important, their sudden introduction breaks the logical progression of the argument. Integrating these definitions into the narrative or placing them in a “Preliminaries” subsection would improve cohesion.

Third, there are minor grammatical issues. In Section 5, the final sentence of the “Efficient training of recurrence” subsection is a fragment, lacking a main verb after listing optimization techniques. Additionally, the use of informal phrases like “on the flip side” in Section 2, while understandable, could be replaced with more precise academic transitions like “Conversely” to maintain a consistent tone.

Finally, the transition between the discussion of the “bank” example and the introduction of Figure 3 is slightly abrupt. The text jumps from the specific example to a general schematic without a clear bridging sentence explaining *why* the schematic is necessary at that specific point. A brief sentence linking the specific failure mode to the general architectural constraint would enhance readability.

Addressing these writing issues will significantly improve the clarity and professional quality of the manuscript without altering its scientific content.

Action items.

- **[writing]** In the Introduction, the sentence ‘Until the transformer... came along, feedforward nets did not seem like a viable approach...’ is historically inaccurate. It implies feedforward nets were never viable, contradicting the use of MLPs. Rephrase to specify that purely feedforward sequence models were limited for long-range dependencies.
- **[writing]** In Section 2, the phrase ‘on the flip side’ is too informal for this context. Replace with ‘Conversely’ or ‘In contrast’ to improve the academic tone and clarity of the logical

transition between the two game examples.

- **[writing]** In Section 3, the standalone paragraph defining ‘Teacher forcing’ and ‘Attractor dynamics’ disrupts the narrative flow. Integrate these definitions into the surrounding text or move them to a ‘Preliminaries’ subsection for better cohesion.
- **[writing]** In Section 5, the final sentence of the ‘Efficient training of recurrence’ subsection is a grammatical fragment. It lists techniques but lacks a main verb (e.g., ‘are proposed’). Add a verb to complete the sentence.